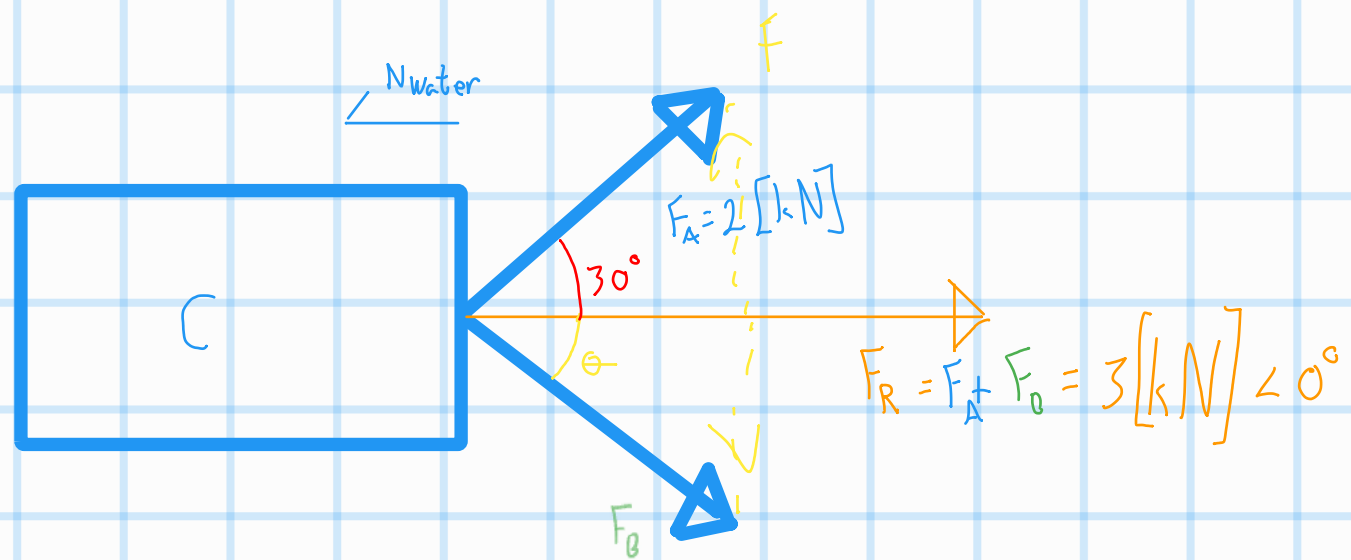


0 pp/ 1

a)



b)

$$F_A = \begin{bmatrix} 2 \cos(30^\circ) \\ 2 \sin(30^\circ) \end{bmatrix}$$

c)

$$F_B = \begin{bmatrix} r \cos \theta \\ r \sin \theta \end{bmatrix} \quad F = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

$$r \sin \theta + 2 \sin(30^\circ) = 0$$

$$r \cos \theta + 2 \cos(30^\circ) = 3$$

$$r = \frac{3 - \frac{\sqrt{3}}{2}}{\cos \theta}$$

$$\left(\frac{3 - \sqrt{3}}{\cos \theta} \right) \sin \theta + 1 = 0$$

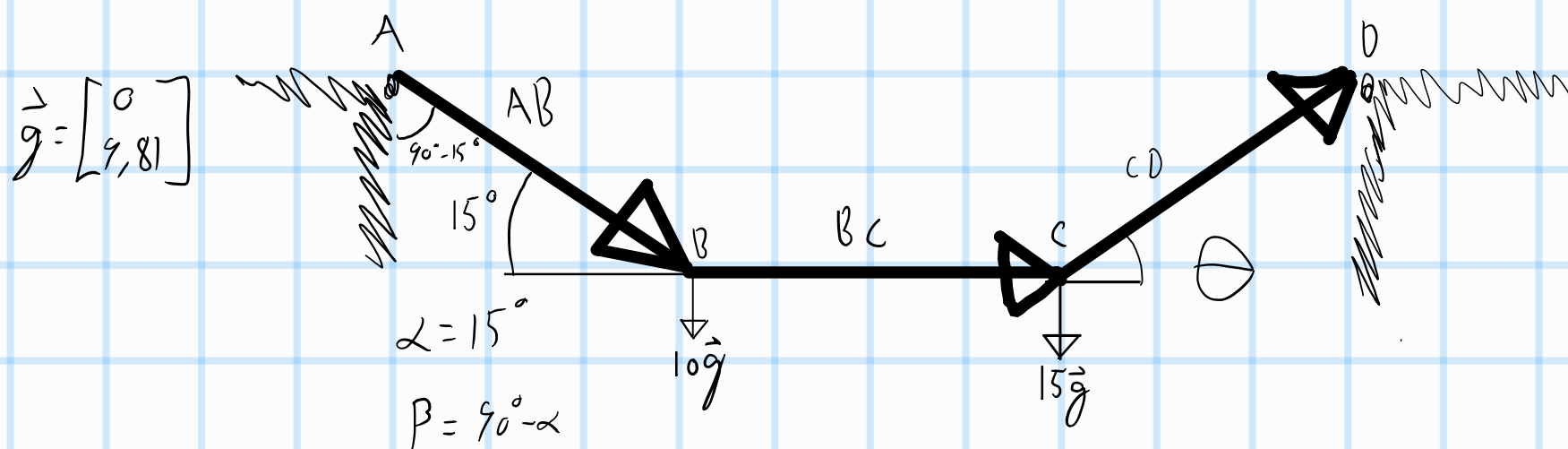
$$F_B = \begin{bmatrix} 3 - \sqrt{3} \\ -1 \end{bmatrix}$$

$$(3 - \sqrt{3}) \tan \theta = -1$$

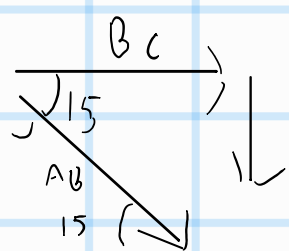
$$\theta = \arctan\left(\frac{-1}{3 - \sqrt{3}}\right) \approx 38,26^\circ$$

$$r = \left(\frac{3 - \sqrt{3}}{\cos\left(\arctan\left(\frac{-1}{3 - \sqrt{3}}\right)\right)} \right) \approx 1,6148$$

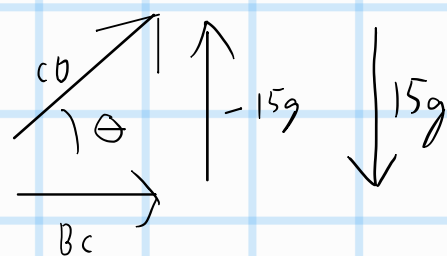
oppz 2



$$g = 9.8$$



$$B: \vec{AB} = 10\vec{g} + \vec{Bc} \Rightarrow 10 \begin{bmatrix} 0 \\ g \end{bmatrix} + \begin{bmatrix} B_c \\ 0 \end{bmatrix} = \begin{bmatrix} AB \cos \alpha \\ AB \sin \alpha \end{bmatrix}$$



$$C: \vec{CD} = 15\vec{g} + \vec{CB} \Rightarrow 15 \begin{bmatrix} 0 \\ g \end{bmatrix} + \begin{bmatrix} CB \\ 0 \end{bmatrix} = \begin{bmatrix} CD \cos \theta \\ CD \sin \theta \end{bmatrix}$$

$$|\vec{CB}| = |\vec{Bc}| \Rightarrow CB = B_c$$

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man begyrr ut fra sine respektable punkt

$$15g = CD \sin \theta$$

$$CB = B_c = CD \cos \theta$$

$$\frac{B_c}{\cos \theta} = CD$$

$$15g = \frac{B_c \sin \theta}{\cos \theta}$$

$$\frac{15g}{B_c} = \tan \theta$$

$$\theta \approx 21.89^\circ$$

$$|\vec{AB}| \approx 379.029 \text{ [N]}$$

$$|\vec{Bc}| = 392.4 \text{ [N]}$$

$$|\vec{CD}| \approx 422.889 \text{ [N]}$$

$$AB \cos \alpha = B_c$$

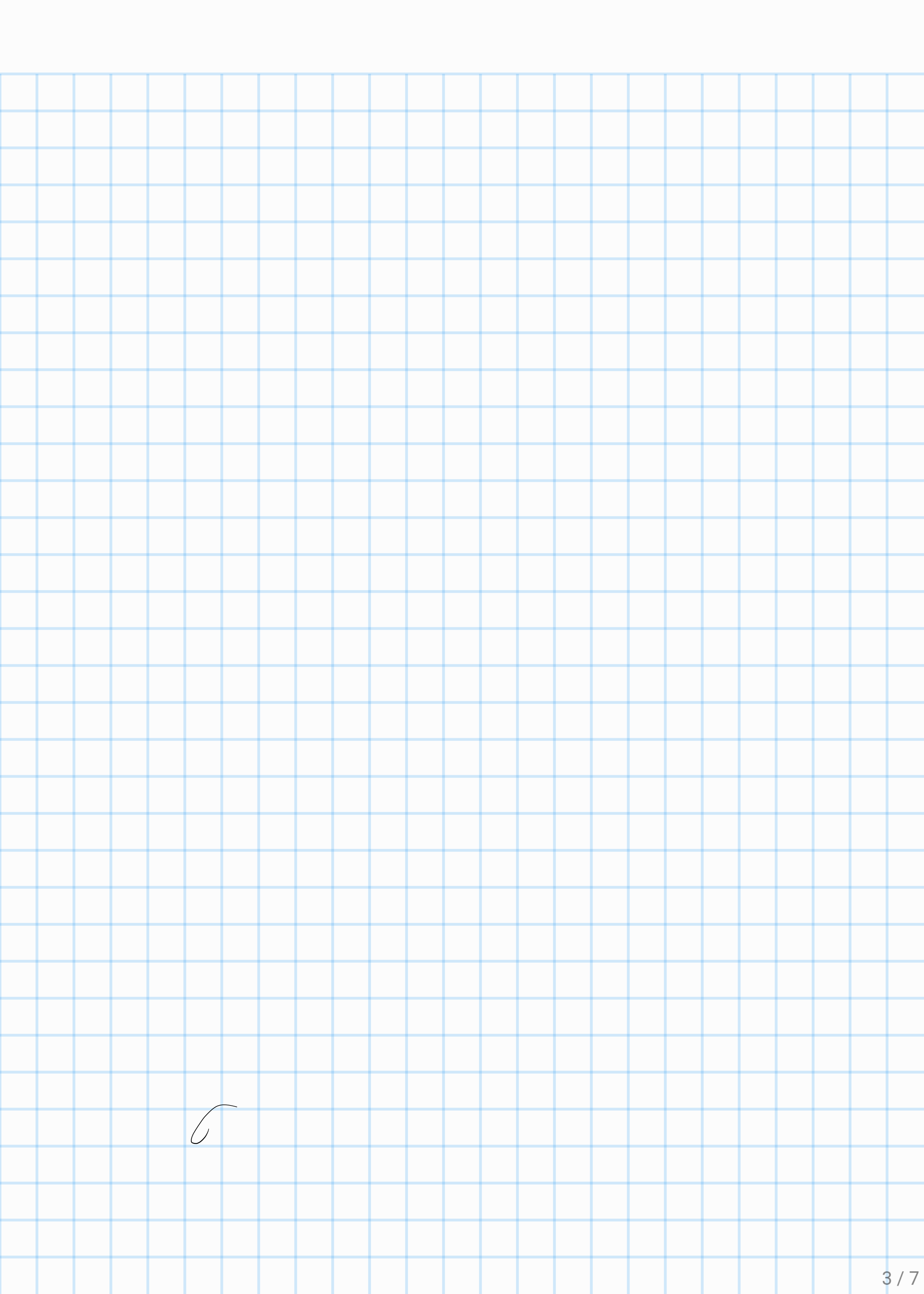
$$10g = AB \sin \alpha \Rightarrow \frac{10g}{\sin \alpha} = AB$$

$$\frac{10g}{\sin \alpha} \cos \alpha = B_c$$

$$AB = \frac{B_c}{\cos \alpha}$$

$$\frac{10g}{\tan \alpha} = B_c$$

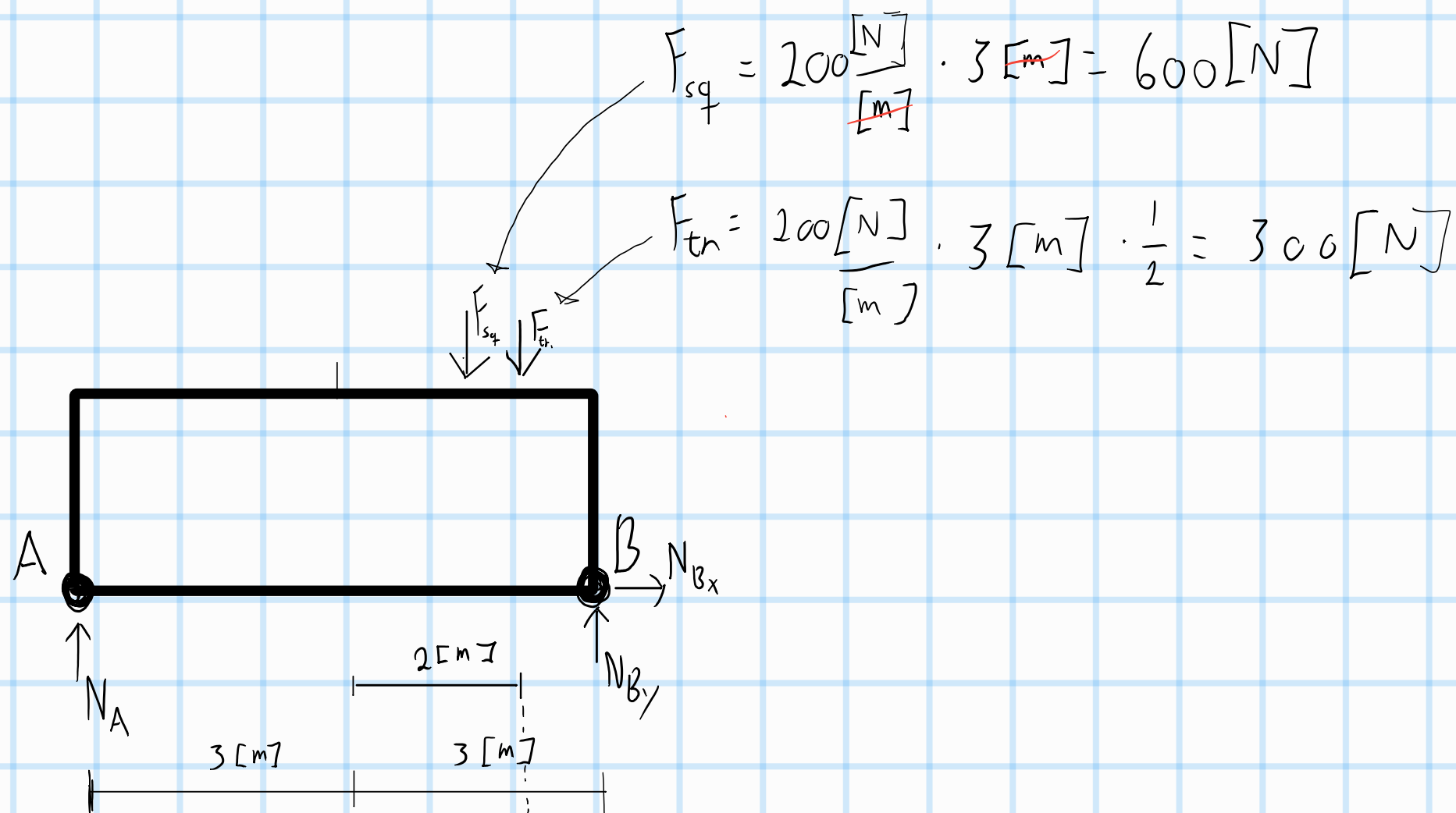
$$\arctan \left(\frac{15g}{\frac{10g}{\tan \alpha}} \right) = \arctan(1.5 \tan \alpha)$$



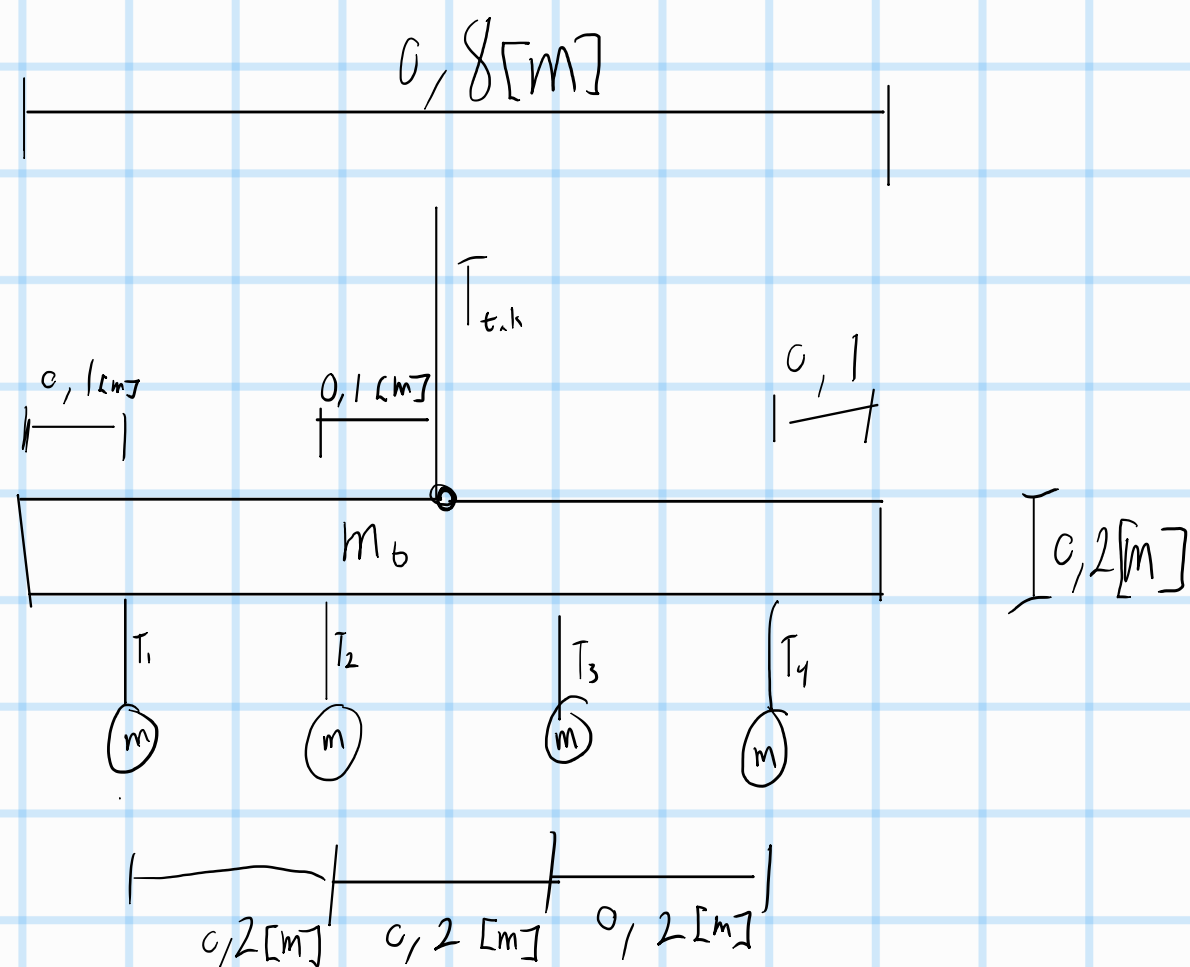
6

Oppg 3

Part 1



Part 2



$$m = 0.5 [kg]$$

$$m_b = 3 [kg]$$

$$\vec{g} = \begin{bmatrix} 0 \\ g \end{bmatrix}, g = 9.81 [m/s^2]$$

$$T_1 = T_2 = T_3 = T_4 = |m \cdot \vec{g}| = 4.905 [N]$$

$$\vec{W}_b = m_b \vec{g} = \begin{bmatrix} 0 \\ 29.43 \end{bmatrix} [N]$$

$$T_{tak} - T_1 - T_2 - T_3 - T_4 - |\vec{W}_b| = 0$$

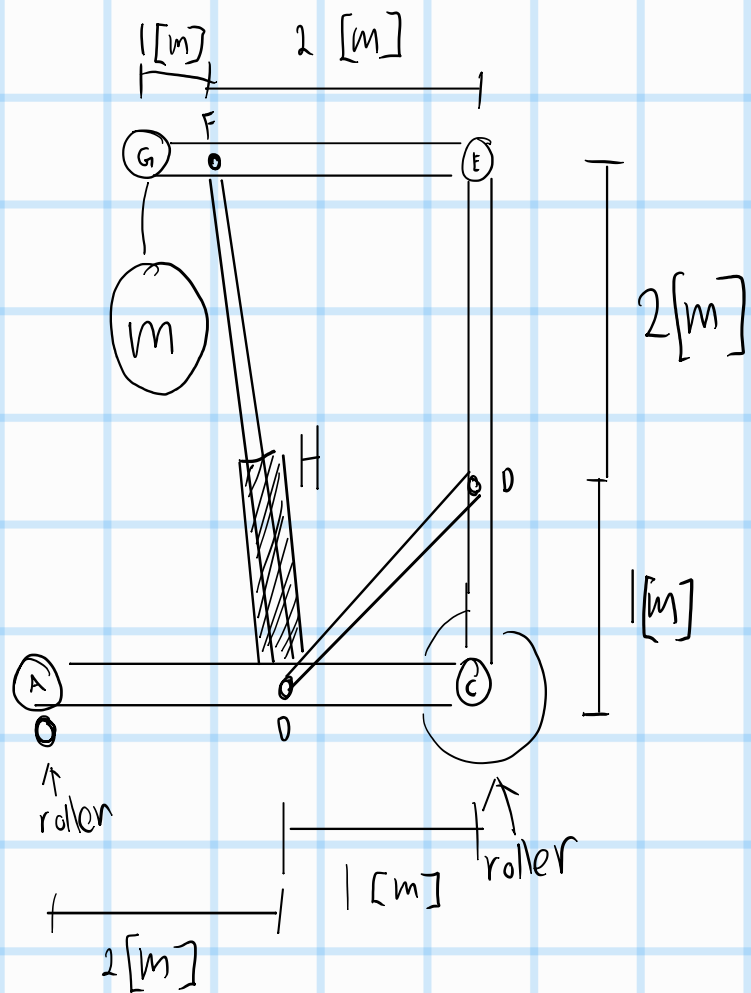
$$T_{tak} = 19.62 + 29.43 = 49.05 [N]$$



088g 4.

using hoist (exercise 6.82 page 179)

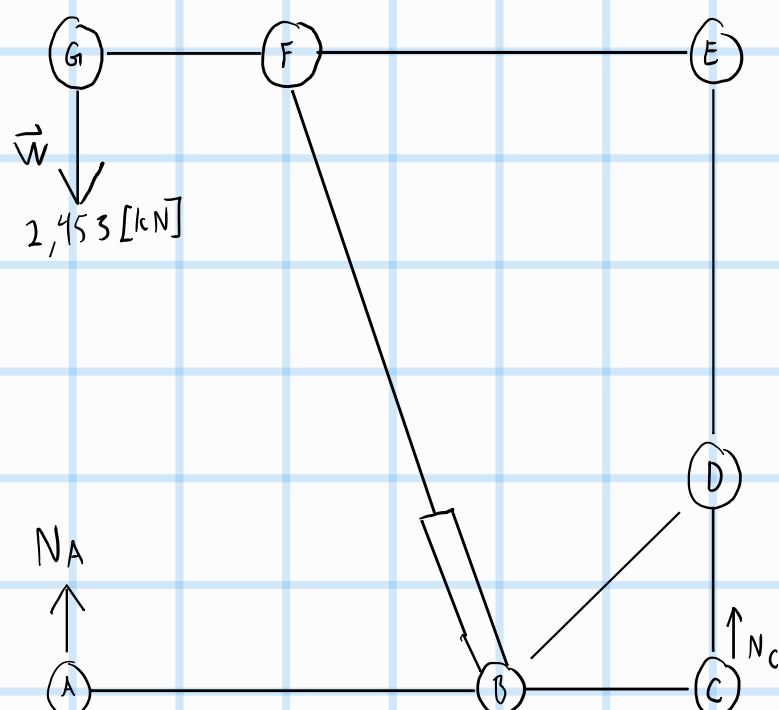
a)



$$P_A = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad P_C = \begin{bmatrix} 3 \\ 0 \end{bmatrix} \quad P_E = \begin{bmatrix} 3 \\ 3 \end{bmatrix} \quad P_G = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

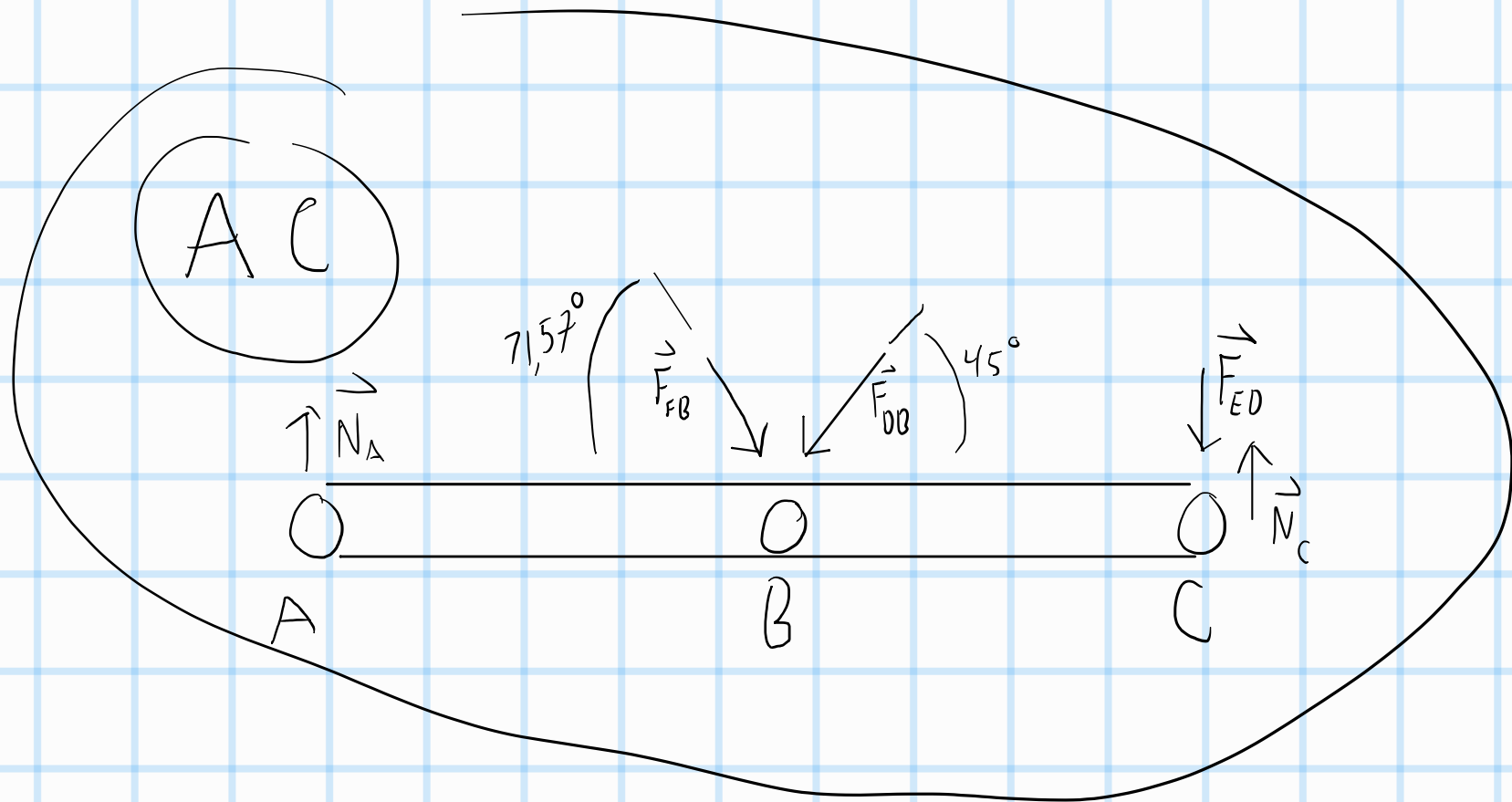
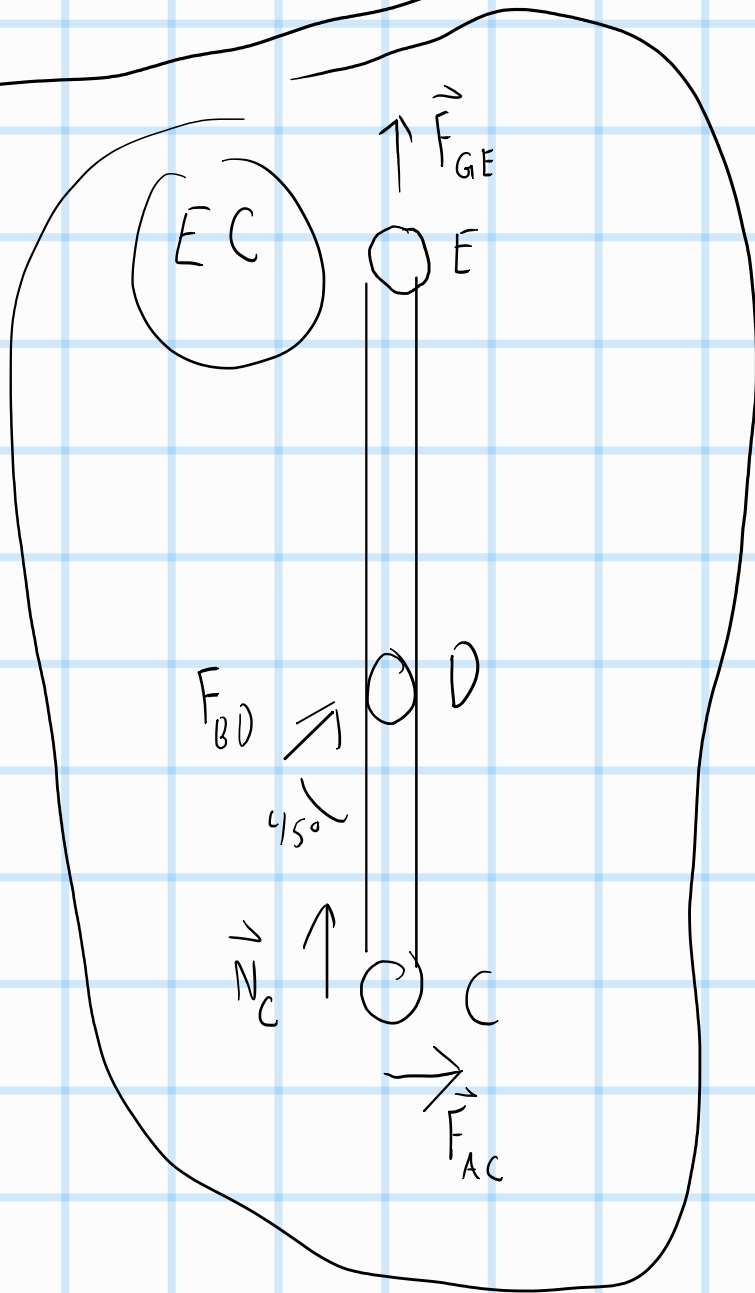
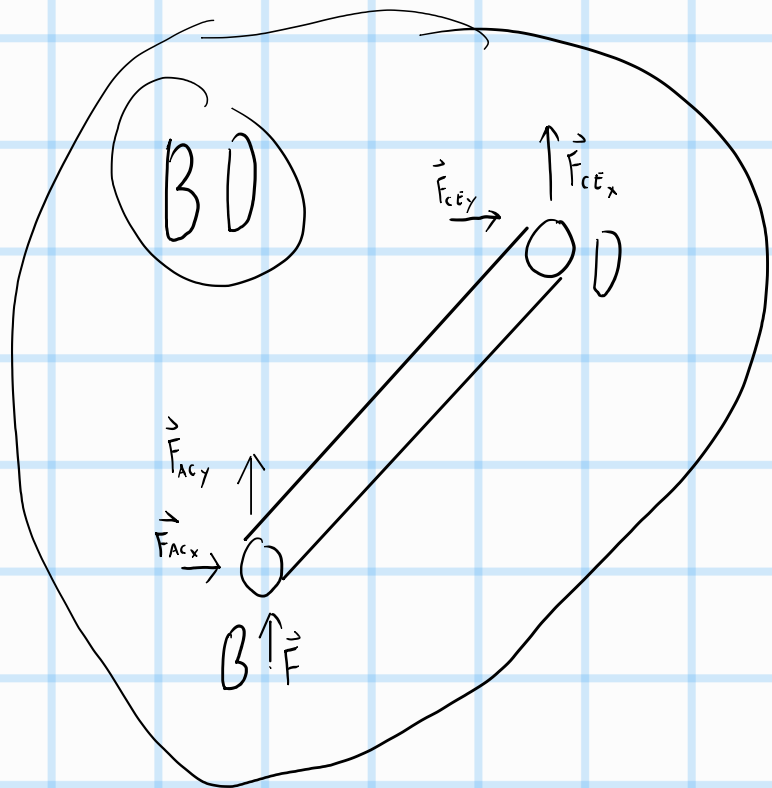
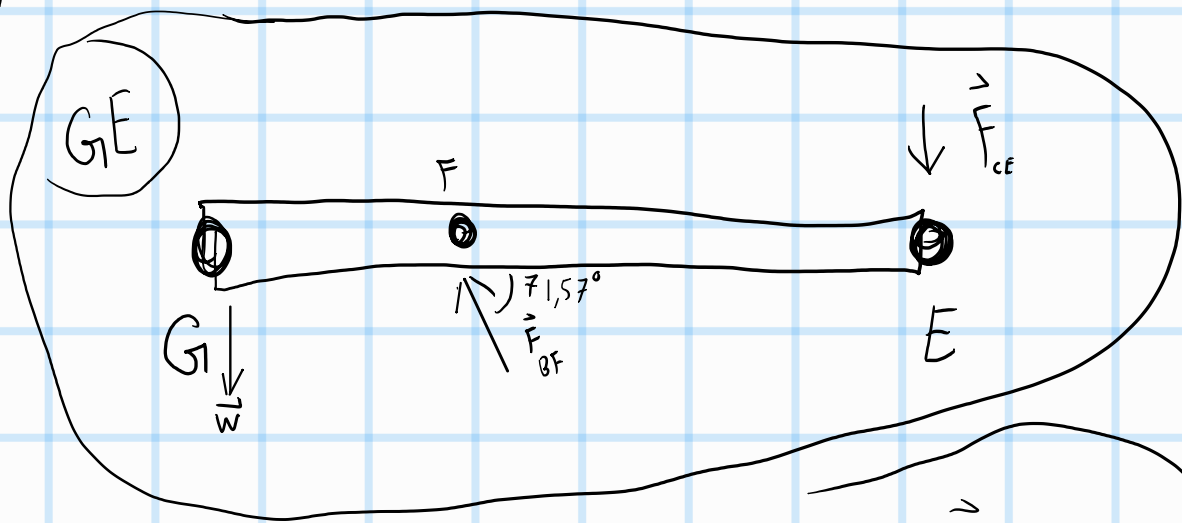
$$P_B = \begin{bmatrix} 2 \\ 0 \end{bmatrix} \quad P_D = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \quad P_F = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad \vec{g} = \begin{bmatrix} 0 \\ g \end{bmatrix}, \quad g = 9,81 \text{ [m/s}^2\text{]}$$

$$m = 250 \text{ [kg]}, \quad \vec{W} = m \vec{g} = 2452,5 \text{ [N]} \approx 2,453 \text{ [kN]}$$



opp g 4

b)



d) real possible weight (m) = 250 [kg]

$$|\vec{F}_G| = 2,45 \text{ [kN]}$$

$$|\vec{F}_{GF}| = 2,45 \text{ [kN]}$$

$$|\vec{F}_{FE}| = 1,74 \text{ [kN]}$$

$$|\vec{F}_{ED}| = 1,74 \text{ [kN]}$$

$$|\vec{F}_{DC}| = 1,74 \text{ [kN]}$$

$$|\vec{F}_{CB}| = 1,23 \text{ [kN]}$$

$$|\vec{F}_{BA}| = 0,82 \text{ [kN]}$$

$$|\vec{F}_{DB}| = 3,48 \text{ [kN]}$$

$$|\vec{F}_{FB}| = 3,88 \text{ [kN]}$$